Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 4 Handout

Tuesday, September 22

Problem 4.1

When making investment decisions, it is helpful to group companies by their size and their market. Let random variable $X \in \{1, 2, \dots\}$ represent the company's revenue in millions of dollars, and random variable $Y \in \{\texttt{local}, \texttt{global}\}$ represent whether the company is local (`local`) or global (`global`). For a company chosen at random, the revenue X is a geometric random variable with parameter $1/2$, i.e., $p_X(x) = 1/2^x$ for $x = 1, 2, \dots$. Moreover, the probability that the company has revenue of x million dollars and it is global is $g(x) = (a^x - 1)$ times the probability that the company's revenue is x and it is local. Assume that a is a known constant, $a > 1$.

(a) Determine joint PMF $p_{X,Y}$.

(b) Determine the probability that a randomly chosen company is local.

(c) All companies are taxed at a rate $\alpha > 0$ of their revenues. In addition, a global company must pay a tax of $c > 0$ million dollars. Assume α and c are known constants. Determine the expected tax a company chosen at random pays.

Problem 4.2

In this problem, we explore indicator variables.

(a) We toss a fair coin n times. Let X_i be the indicator random variable that corresponds to coin i landing heads. Let Y be the total number of tosses that turned up heads. Determine $\mathbb{E}[X_i]$ and $\mathbb{E}[Y]$.

(b) A monkey types 1,000,000,000 characters using the 26 letters of the alphabet. The monkey selects each character independently of all other strokes and is equally likely to choose any of the 26 letters. Determine the expected number of occurrences of the word `caesar`.

(c) Consider a standard 52-card deck that has been shuffled. Determine the expected number of times when a king in the deck is followed by another king. For example, if all four kings are next to each other, then we'd say that there are 3 kings that are followed by another king.